

Real-Time Automatic Leaf Detection System on Water Surface using YOLO v8

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Abstract— Outdoor ponds and aquariums are often exposed to dust and fallen leaves, which can cause water pollution and filter clogging, potentially harming fish and degrading water quality. This paper presents a real-time automatic leaf detection and alert system on the water surface for outdoor ponds and aquariums. The system uses a GoPro camera mounted on a floating robotic platform and connected to a Raspberry Pi to capture images of the water surface. The captured images are transmitted to a computer, where YOLOv8-based object detection are performed using Google Colab. When large leaves are detected, the system sends an alert notification to the user via the LINE application. The experimental results show that the system can detect fallen leaves with an accuracy of approximately 80%. The purpose of this system is to help pond and aquarium owners maintain clean water, prevent water quality degradation, and reduce potential harm to aquatic life by providing early alerts.

Keywords—IoT, Raspberry Pi, YOLO, Object Detection, Pond and Aquarium Monitoring, Alert System.

I. INTRODUCTION

Outdoor ponds and aquariums are widely used for fish keeping, aquaculture, and decorative purposes. These aquatic environments play an important role in maintaining the health and well-being of fish and other aquatic organisms. However, such environments are often exposed to external factors, including dust, insects, and fallen leaves from surrounding areas. These foreign objects can accumulate over time and significantly affect water cleanliness and overall water quality.

Among these factors, fallen leaves are considered one of the major causes of water pollution and filter clogging in outdoor ponds and aquariums. When leaves fall into the water and are not removed promptly, they undergo a natural decomposition process. This process releases organic matter and contaminants into the water, leading to a decline in water quality. As a result, dissolved oxygen levels may decrease, and harmful substances may increase, which can negatively impact fish health and other aquatic life. In severe cases, this may lead to fish stress or mortality.

In addition, conventional cleaning and monitoring methods rely mainly on manual inspection by the owner. This approach is not efficient, as it cannot provide continuous monitoring or real-time detection. In many situations, fallen leaves may remain unnoticed for a long period, allowing water conditions to deteriorate before any corrective action is taken. Therefore, there is a need for an automated system that can continuously monitor the aquatic environment and provide timely alerts.

To address this problem, this paper proposes an automatic leaf detection and alert system for ponds and aquariums. The proposed system utilizes a GoPro camera mounted on a floating robotic platform and connected to a Raspberry Pi to capture images of the water surface at regular intervals. These images are processed using image processing and AI-based object detection techniques to identify fallen leaves.

By providing continuous monitoring and early detection, the proposed system enhances traditional maintenance methods and improves the efficiency of pond and aquarium management. It helps prevent water quality degradation, reduces the risk of harm to aquatic life, and supports a cleaner and healthier aquatic environment.

II. RELATED THEORIES

This section presents the fundamental theories and technologies used in this study. The proposed system integrates object detection, deep learning algorithms, image processing techniques, and Internet of Things (IoT) concepts to achieve accurate and real-time detection of fallen leaves in aquatic environments.

A. Object Detection

Object detection is a computer vision technique used to identify and localize objects within an image or video. It combines classification, which determines the object category, and localization, which identifies the position of the object within the frame. This technique has been widely applied in various domains such as surveillance systems, autonomous vehicles, and environmental monitoring due to its ability to analyze visual data efficiently.

In this project, object detection is applied to identify fallen leaves on the water surface. The system detects the presence and position of leaves within each captured frame.



Figure 1: Example of Object Detection using Bounding Box

B. YOLOv8 Algorithm

YOLOv8 (You Only Look Once version 8) is a state-of-the-art deep learning model designed for real-time object detection. It provides a strong balance between high detection accuracy and fast processing speed, making it suitable for time-sensitive applications. In this study, YOLOv8 is selected due to its robustness in handling challenging water surface environments, where light reflection, refraction, and dynamic movement can affect object visibility.

The detection process of YOLOv8 begins with an input image captured from the camera, such as images of leaves on the water surface. This image is then passed through the CNN Backbone, which performs feature extraction by identifying important visual characteristics such as edges, shapes, textures, and colors. After that, the extracted features are processed in the Neck, where information from multiple scales is combined, allowing the model to understand both overall structures and fine details of the objects.

Next, the Detection Head analyzes these features to determine what objects are present in the image, their locations, and their categories. Finally, the Output is generated in the form of bounding boxes, class labels (such as mango leaf, tamarind

leaf, or yellow elder leaf), and confidence scores indicating the reliability of the detection. This process enables accurate and real-time leaf detection, which can be further used for decision-making and alert notification in the system.

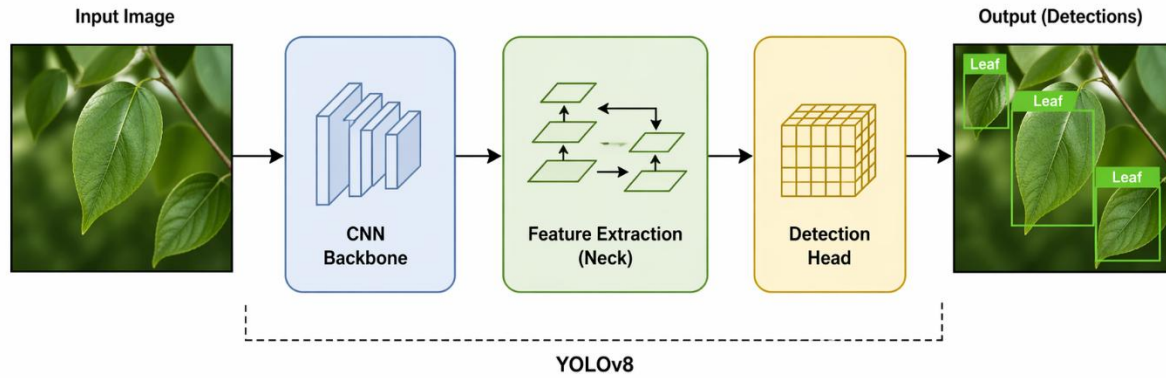


Figure 2: YOLOv8 Detection Framework

C. Internet of Things (IoT)

The Internet of Things (IoT) enables communication and data exchange between devices through network connectivity. It allows systems to be monitored and controlled remotely, making it a key component in modern smart systems.

In this project, IoT is used to deliver real-time alerts to the aquarium or pond owner when fallen leaves are detected. The Raspberry Pi functions as a communication unit, sending notifications to external devices such as smartphones or messaging platforms. This enables timely user response, helping maintain water quality and ensuring a healthier aquatic environment.

D. Data Collection

Dataset preparation and annotation are essential steps in training an object detection model. In this study, images of three types of leaves (tamarind, yellow elder, and mango) are collected from natural environments under various conditions, including different lighting, angles, and backgrounds. The dataset is designed to represent real-world conditions on the water surface, including variations in lighting, leaf orientation, and background complexity. This ensures that the trained model can generalize well in practical environments.

Each image is annotated using bounding box techniques, where the location and class of each leaf is labeled. The annotation process is performed using tools such as Roboflow, and the labeled dataset is used to train the YOLOv8 model. A well-prepared and diverse dataset improves detection accuracy and model robustness in real-world conditions.

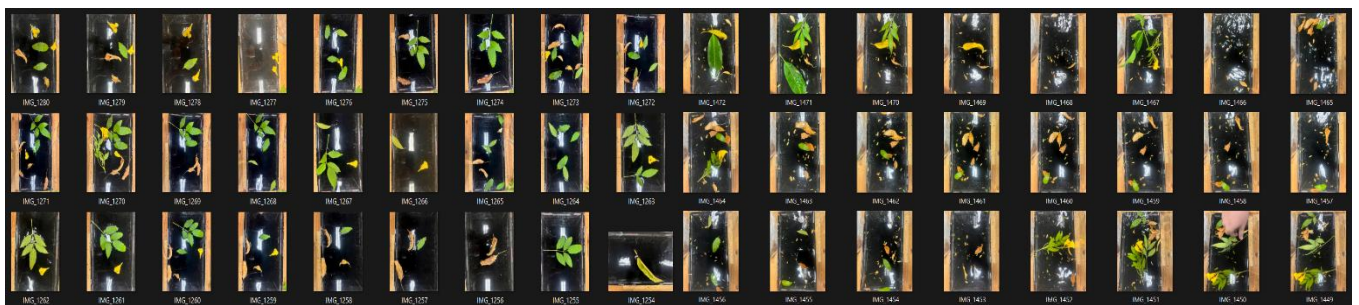


Figure 3. Sample images from the dataset used for leaf detection

III. METHODOLOGY

In the methodology section describes the overall methodology of the proposed system, including data collection, model training, system implementation, and decision-making processes for leaf detection and alert generation.

A. System Overview and Hardware Architecture

The proposed system consists of multiple hardware components integrated to perform real-time leaf detection on the water surface in an aquatic environment. The system includes a Raspberry Pi, Jetson Nano, GoPro Hero 11 camera, power supply modules, and additional robotic components mounted on a floating platform. This design aligns with the project scope, which focuses on detecting and classifying fallen leaves on the water surface and generating alerts based on their size and potential impact on water quality.

The YOLOv8 model is executed on a remote computing platform (Google Colab) using a computer. The Jetson Nano is used for controlling the movement of the robotic platform, including water pump operation.. The GoPro camera is mounted on a floating robotic platform to capture images of the water surface, ensuring continuous monitoring of leaf accumulation. The captured images are transmitted directly to the Jetson Nano, where the system detects and classifies different types of leaves according to the defined categories (small and large leaves). This real-time processing enables the system to make immediate decisions based on the detection results.

The system is powered by a 12V DC battery, with a step-down converter used to provide a stable 5V supply for the Raspberry Pi and other components. A solar panel is installed on top of the robot to support energy sustainability and extend operational time. Additionally, multiple water pumps are used to control the movement of the floating robot within the pond, while sensors such as RPLidar assist in navigation and obstacle detection. This hardware architecture enables the system to operate autonomously in real-world aquatic environments, ensuring efficient monitoring, adaptive movement, and timely response to potential water quality issues.



Figure 4: Overall system architecture of the proposed leaf detection system

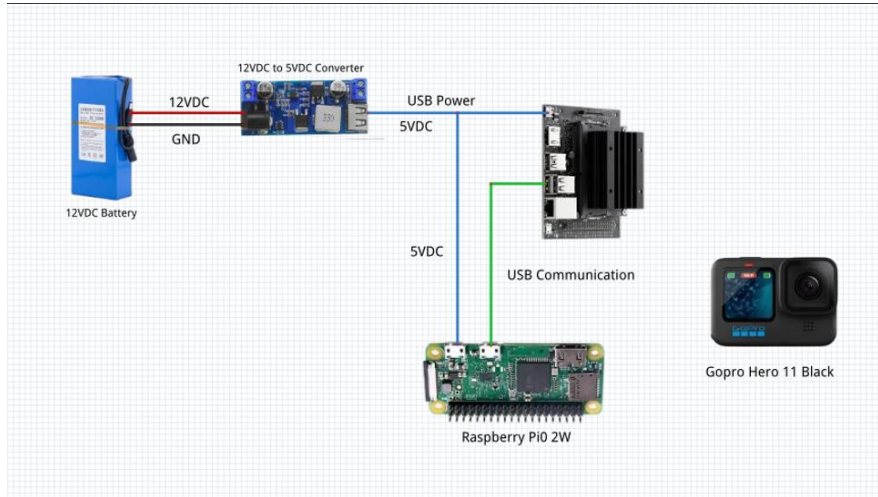


Figure 5: Overall Hardware architecture of the proposed leaf detection system

B. Data Collection and Dataset Preparation

The dataset used in this study was collected from real-world leaf samples obtained from natural environments. Three types of leaves were considered, including tamarind leaves, yellow elder leaves, and mango leaves. To ensure diversity and improve model performance, both dry (brown) and fresh (green) leaves were included in the dataset.

A total of 1000 images were collected for training and testing purposes. Specifically, 150 images were collected for each individual leaf type (tamarind, yellow elder, and mango), resulting in 450 single-class images. In addition, 550 images containing multiple leaf types within a single frame were captured to improve the model's ability to detect and classify multiple objects simultaneously in real-world scenarios.

All images were annotated using bounding box techniques, where each leaf was labeled according to its corresponding class. The annotation process was performed using Roboflow, and the dataset was exported in a format compatible with the YOLOv8 model. This diverse and well-annotated dataset plays a crucial role in improving detection accuracy and robustness under varying environmental conditions.



Figure 6. Sample dataset images of Mango leaf



Figure 7. Sample dataset images of Yellow Elder leaf



Figure 8. Sample dataset images of Tamarind leaf

C. Model Training Using YOLOv8

The prepared dataset is used to train the YOLOv8 object detection model. During the training process, the model learns to recognize and classify different types of leaves based on their visual features. The training aims to achieve high detection accuracy and reliable classification performance.

During the training process, the model loss for both training and validation datasets shows a decreasing trend, indicating that the model is learning effectively. Although the initial training stages may exhibit fluctuations, the overall loss converges as the training progresses. The final trained model achieves stable performance with improved accuracy, demonstrating its capability to generalize well to unseen data.

D. Decision-Making and Alert System

After detecting the leaves, the system performs a decision-making process based on the classification results. If a small leaf (tamarind leaf) is detected, the system records the detection but does not generate an alert. However, if a large leaf (yellow elder or mango leaf) is detected, the system immediately triggers an alert.

The alert is sent to the user via LINE application using IoT communication, allowing timely action to remove the leaf. This mechanism ensures that potential water quality issues are addressed promptly and efficiently.

The alert system was initially developed and tested in a prototype stage, where some instability in notification delivery was observed. However, after further improvements and optimization, the system is now able to reliably send real-time notifications via the LINE application. This enhancement ensures consistent communication between the system and the user.

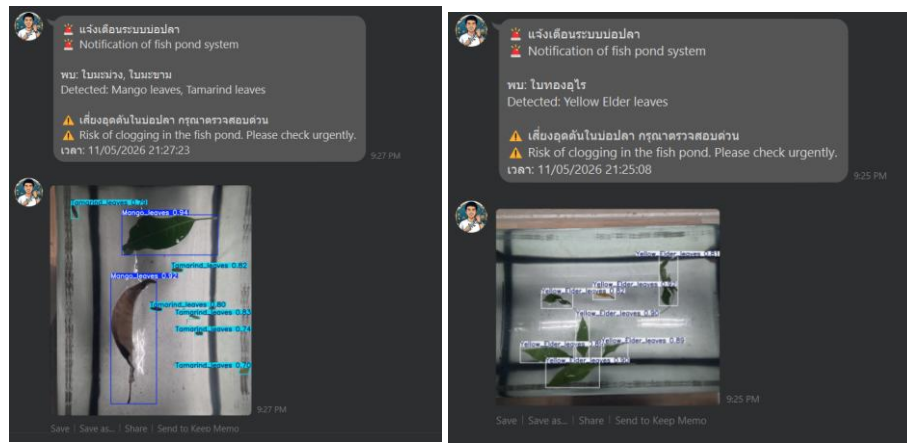


Figure 9: Example of leaf detection results and alert notifications sent via the LINE application

E. System Workflow

The overall system workflow begins with the GoPro camera capturing one image from the water surface at regular intervals of every five minutes, instead of continuous image acquisition. The Raspberry Pi then transmits the captured image to a computer for processing. The YOLOv8 model, running on Google Colab, performs object detection to identify leaf types.

Based on the detection results, the system classifies leaves into small or large categories. If a large leaf is detected, a notification is sent to the user via the LINE application. Meanwhile, the Jetson Nano controls the robot movement through water pumps, enabling the robot to navigate and position itself effectively within the aquatic environment.

IV. IMPLEMENTATION

A. Hardware Implementation

The hardware implementation of the proposed system consists of multiple components working together to enable real-time leaf detection and monitoring in aquatic. The Jetson Nano is used for controlling the movement of the robotic platform, including the operation of water pumps. The YOLOv8 model is executed on a remote computing platform (Google Colab) using a computer for object detection and performing real-time image processing. A GoPro Hero 11 Black camera is used to capture high-quality images from the water surface for accurate detection. In addition, a Raspberry Pi functions as a secondary controller, handling communication and IoT-based alert transmission to notify users when large leaves are detected.

The system is powered by a 12V DC battery, which is regulated using a step-down converter to provide a stable 5V supply for low-power components. A solar panel is integrated to enhance energy sustainability and extend operation time. The robotic platform is designed to float on the water surface and is equipped with four water pumps for movement and positioning. Additionally, an RPLidar sensor is included to support environmental awareness and navigation. All hardware components are integrated into a compact floating structure, enabling continuous and autonomous operation in real-world aquatic environments.

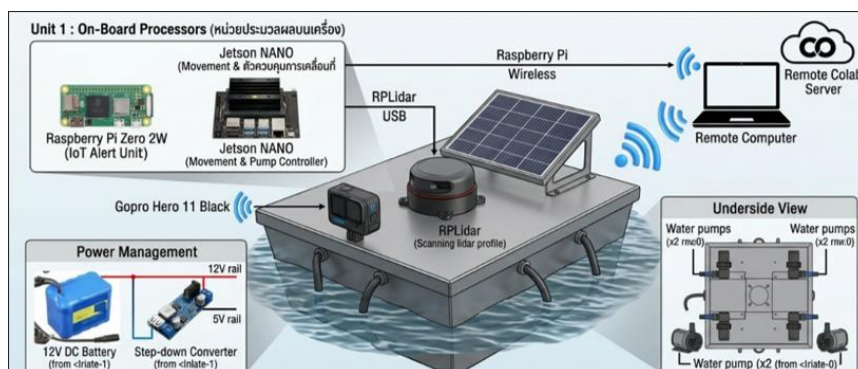


Figure 10: On-Board Hardware Components of the Proposed Floating Robotic System

B. Software Implementation

The software implementation of this system involves dataset preparation, model training, and deployment of the trained model for real-time detection. The dataset consists of three types of leaves: tamarind leaves, yellow elder leaves, and mango leaves. These leaves are categorized into two groups: small leaves (tamarind) and large leaves (yellow elder and mango), where only large leaves trigger alert notifications.

The dataset is collected from natural environments and includes both dry (brown) and fresh (green) leaves. For each leaf type, approximately 300 images are captured under various conditions, including different orientations, lighting conditions, and backgrounds. Additionally, around 500 images are collected containing mixed leaf types in a single frame to improve the model's ability to detect multiple objects simultaneously.

All images are annotated using Roboflow, where bounding boxes and class labels are assigned to each leaf. The annotated dataset is then exported in a format compatible with YOLOv8 and used for training in Google Colab using Python scripts. The YOLOv8 model is trained to recognize and classify the three types of leaves with high accuracy.

After achieving satisfactory performance, The trained YOLOv8 model is executed on Google Colab, where images transmitted from the Raspberry Pi are processed for object detection. The system processes input images captured from the camera and performs continuous detection in a real-time loop.

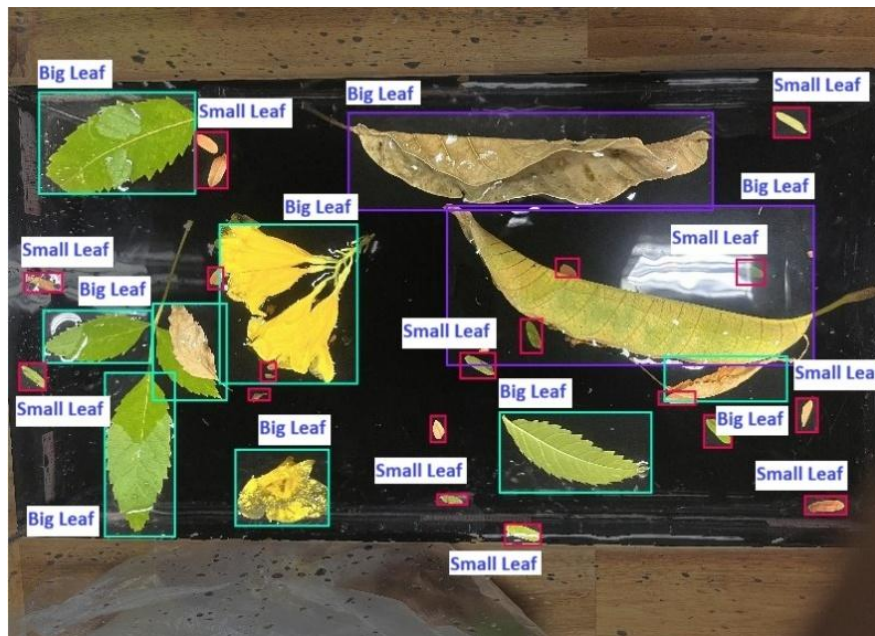


Figure 11: Leaf classification into small and large categories used in this study

C. System Integration

The system integration describes how all hardware and software components are connected to operate as a complete real-time leaf detection system. The process begins with the GoPro camera, which continuously captures images from the water surface. These images are then transmitted to a computer, where the YOLOv8 model running on Google Colab performs object detection to identify and classify different types of leaves.

Once the detection process is completed, the system performs decision-making based on the classification results. If a small leaf (tamarind leaf) is detected, the system records the information without generating an alert. However, if a large leaf (yellow elder or mango leaf) is detected, the system sends the detection result to the Raspberry Pi. The Raspberry Pi then processes this information and triggers an alert notification through IoT communication to the user's device, enabling immediate action.

The system captures one image every five minutes instead of continuous image acquisition, reducing computational load and power consumption while maintaining effective monitoring.

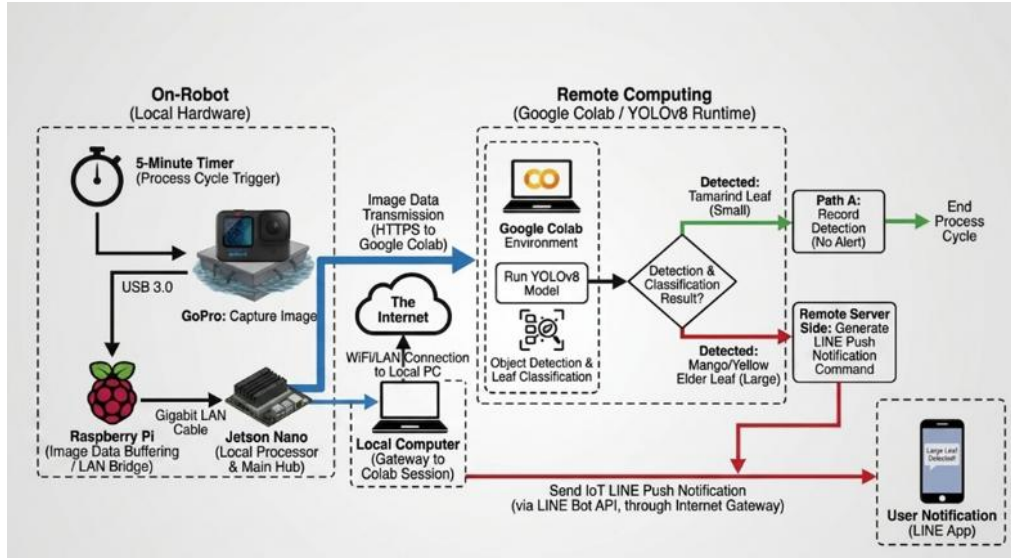


Figure 12: Functional Integration and data flow diagram of the cyclical leaf detection process, illustrating YOLOv8 decision-making with a local multi-stage hardware path, and a direct remote-to-LINE-App

D.Alert and Communication System

The alert and communication system is designed to notify users when large leaves are detected on the water surface. After the object detection process is completed on a computer using the YOLOv8 model running on Google Colab, the detection results are sent to the Raspberry Pi for further processing. The Raspberry Pi acts as the communication unit, responsible for handling data transmission and generating alerts.

When a large leaf (such as a mango or yellow elder leaf) is detected, the system immediately triggers a notification to the user through IoT-based communication platforms such as the LINE application. In contrast, when a small leaf (tamarind leaf) is detected, the system only records the detection without sending an alert, reducing unnecessary notifications.

The system captures one image every five minutes and processes it accordingly, enabling efficient monitoring while reducing computational load and power consumption. This approach ensures reliable system operation and timely notification to the user.

V. RESULT

This section presents the performance evaluation of the YOLOv8 model in detecting fallen leaves on the water surface, as well as the operational results of the alert system via the LINE application. The experiment is divided into two main parts: the evaluation of the model's accuracy and the testing of the system's integration in simulated scenarios.

A. Model Performance

The learning progression of the YOLOv8 model over 100 epochs is illustrated through the Model Accuracy and Model Loss graphs, as shown in Figure 13.

The Model Accuracy graph demonstrates a continuous upward trend in the mean Average Precision (mAP50) on the validation set, which begins to stabilize and converge after the 40th epoch, reaching a peak accuracy of approximately 0.837 (84%). Simultaneously, the Model Loss graph reveals a steady decline in both training and validation loss. This parallel decrease in loss values is a strong indicator that the model effectively learned the robust features of the leaves without suffering from severe overfitting. The convergence of these metrics validates the model's reliability and its readiness for real-world deployment in the automated leaf detection system.

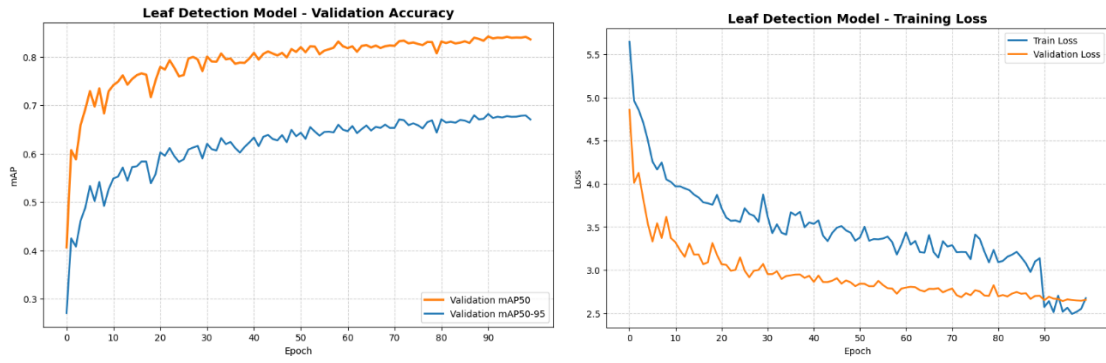


Figure 13: Validation accuracy and Training loss graph

The performance in classifying leaf types was evaluated using a confusion matrix, as illustrated in Figure 13. The experimental results indicate that the model achieves exceptionally high accuracy in detecting large leaves, successfully identifying mango leaves with 90% accuracy and yellow elder leaves with 84% accuracy.

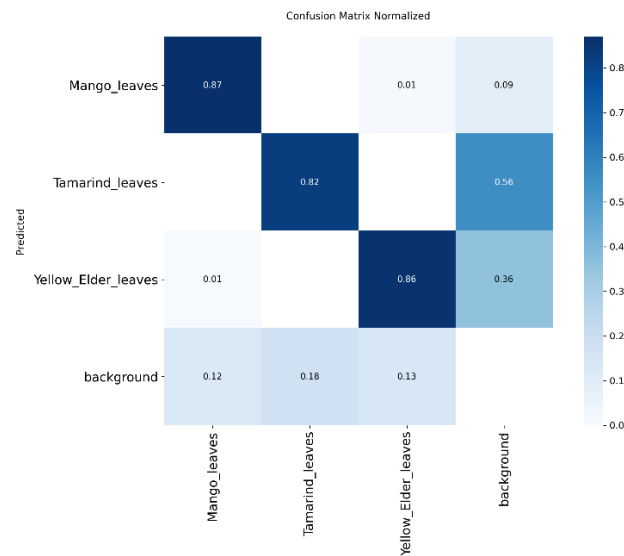


Figure 14: Normalized Confusion Matrix of the YOLOv8 model showing the detection accuracy across three leaf classes and background.

The normalized confusion matrix illustrates the high efficiency of the YOLOv8 model in identifying the target objects. According to the experimental results:

- Mango leaves (large category) were detected with an accuracy of 87%.
- Yellow Elder leaves (large category) were detected with an accuracy of 86%.
- Tamarind leaves (small category) achieved a detection accuracy of 82%.

Regarding misclassifications, the model identified 18% of actual tamarind leaves as background (False Negatives), likely due to their miniature size and color degradation. Additionally, when the model incorrectly detected background elements as objects, 56% of those instances were misclassified as tamarind leaves (False Positives), often triggered by water ripples or shadows. Despite these variations, the system remains reliable as the logic is programmed to suppress notifications for small debris like tamarind leaves.

B. System Integration and Alert Notification

The system was tested in a real-time environment through a Graphical User Interface (GUI) developed using CustomTkinter. The system smoothly acquired images from the camera, processed the detection, and accurately rendered the bounding boxes, as shown in Figure 15.

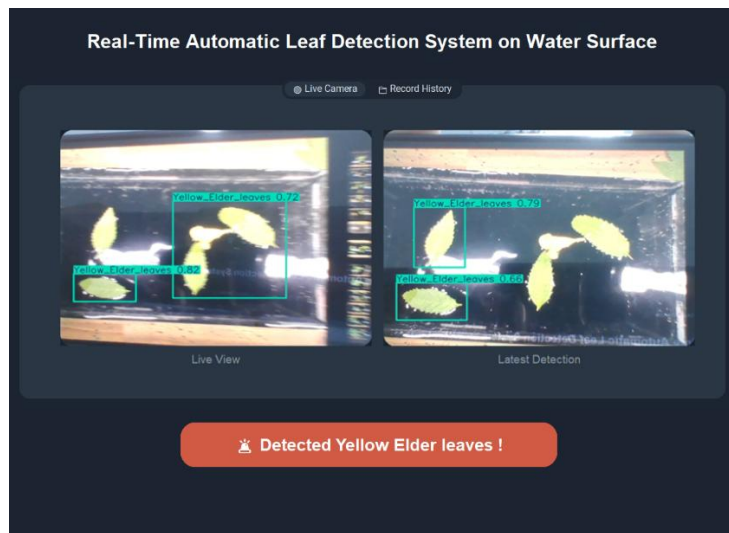


Figure 15: The Graphical User Interface (GUI) displaying real-time leaf detection and current system status.

Regarding the smart alert system, the decision-making logic functioned exactly as designed. When the system detected large leaves (mango or yellow elder) that pose a high risk of clogging, it immediately captured the image and transmitted a bilingual (Thai/English) alert notification to the user via the LINE Messaging API, as illustrated in Figure 16. Conversely, if only tamarind leaves were detected, the system displayed the detection status solely on the GUI for monitoring purposes, without sending disruptive messages to the LINE application. This mechanism effectively mitigates the issue of alert fatigue and ensures that users are only notified of critical events.

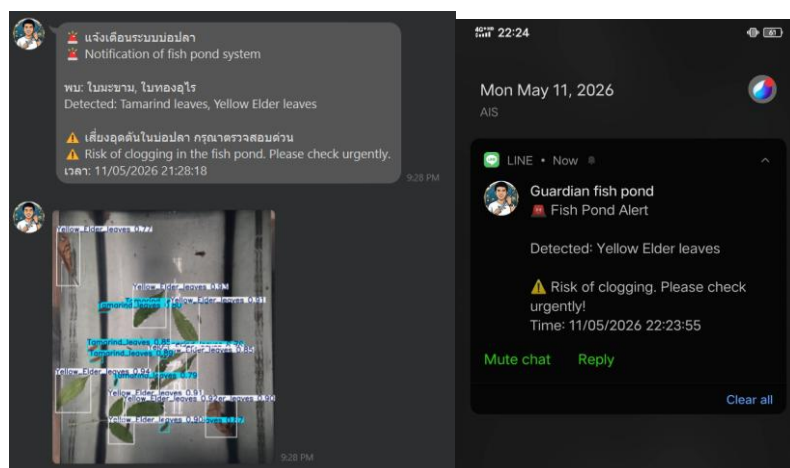


Figure 16: Example of real-time alert notification sent via the LINE application, including the detected image and bilingual warning message.

C. Qualitative Detection Results

To further demonstrate the robustness and practical performance of the trained YOLOv8 model, additional real-world detection tests were conducted using actual images captured from the water surface.

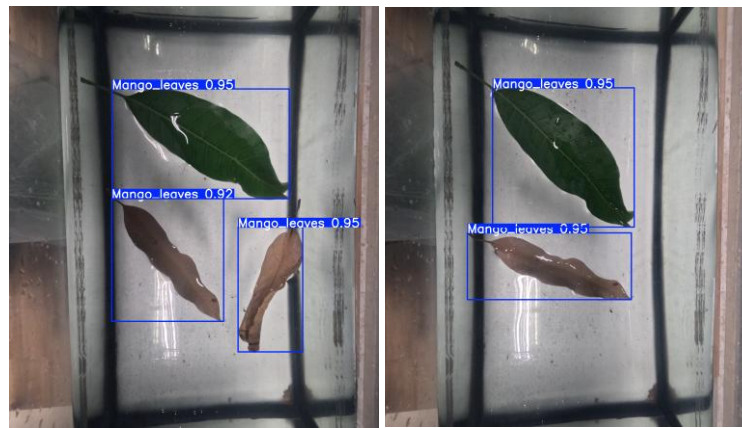


Figure 17: YOLOv8 detection result of mango leaves on the water surface.

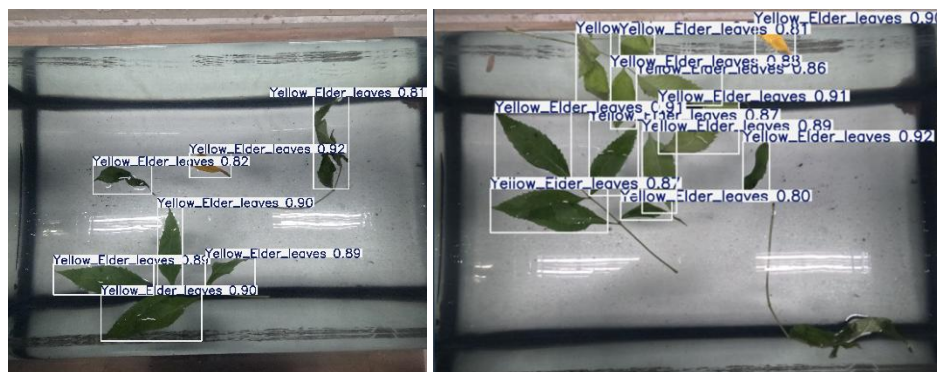


Figure 18: YOLOv8 detection result of yellow elder leaves on the water surface.

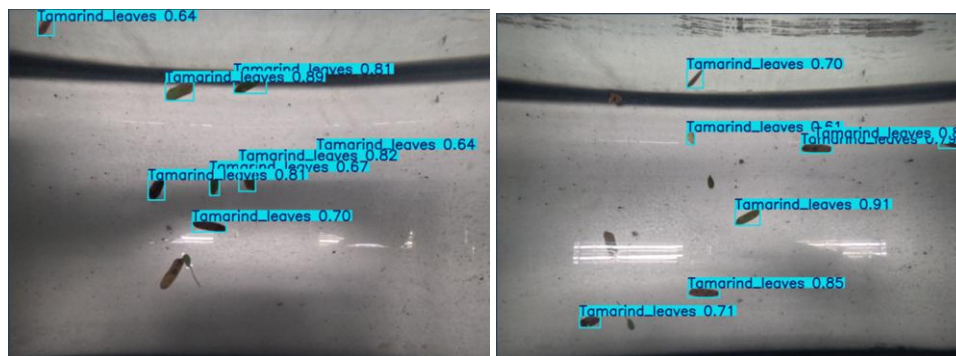


Figure 19: YOLOv8 detection result of tamarind leaves (small leaves).

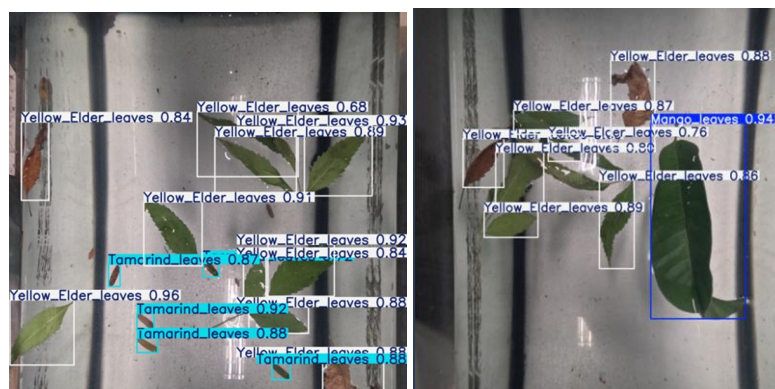


Figure 20: YOLOv8 detection results in a complex scene with multiple mixed leaf types.

As shown in Figures 17–20, the model successfully detected and localized the leaves with high confidence scores. Mango leaves were typically detected with confidence between 0.85–0.95, while yellow elder leaves ranged from 0.75–0.89. The model also demonstrated good performance in distinguishing small tamarind leaves despite their size and lower contrast against the water background. Tight bounding boxes and minimal false detections were observed even under challenging conditions such as water ripples and specular reflections.

D. Conclusion and future work

This study presents an automatic leaf detection and alert system for water surface environments using the YOLOv8 model. The system can detect and classify different types of leaves and categorize them into small and large groups. The experimental results show that the system achieves approximately 84% accuracy and can successfully generate alert notifications via the LINE application when large leaves are detected.

For future work, the system can be extended to detect other floating objects such as plastic waste, bottles, and cans to enhance its applicability in environmental monitoring. Additionally, reducing the image capture interval from five minutes to one minute can improve detection reliability for objects that may sink quickly. Further improvements may include optimizing model performance, increasing accuracy under varying conditions, and enhancing system robustness for long-term real-world deployment.

VI. REFERENCES

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